Katherine Piper

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Education

Carnegie Mellon University | Pittsburgh, Pennsylvania — August 2024 – Present
Doctor of Philosophy | Computational Mechanics — Est. April 2029
Master of Science | Computational Mechanics — Awarded December 2025

California Institute of Technology | Pasadena, California — September 2019 – June 2024
Bachelor of Science | Mechanical Engineering

Research Experience

Carnegie Mellon University | Pittsburgh, Pennsylvania
Graduate Researcher — August 2024 – Present
- Constrained optimization machine learning methods for physics and engineering problems

DEVCOM Army Research Laboratory | Aberdeen, Maryland
Research Associate (Advisors: Jarek Knap and Kevin LoGrande) — Summer 2026
- Advancing Diffusion Maps Surrogates with Constrained Optimization

NASA Jet Propulsion Laboratory | Pasadena, California
Academic Part Time Employee — October 2023 – August 2024
Year Round Intern — May 2023 – September 2023
- Primary Lab: Additive Manufacturing Center (AMC)
- Secondary Lab: Two-Phase Thermal Technology Laboratory
- Additively manufactured heat pipe design
- Supervised testing of additively manufactured thermal control systems
- Methods lead to 100 times cost reduction and 50% performance improvement over state-of-the-art
- Two NASA New Technology Reports and US Patent submitted

California Institute of Technology | Pasadena, California
Summer Undergraduate Research Fellowship — June 2022 – June 2024
Research Advisor – Vinamra Agrawal, Los Alamos National Laboratory
- Implementation of a model for ductile fracture in functionally graded additively manufactured metals

Awards

CEE Research Spotlight: 1 of 4 Featured Posters — August 2026
CMU CEE Fall 2026 Poster Competition

Journeyman Fellowship — June-August 2026
DEVCOM Army Research Laboratory

Grand Prize: Sustainable Impact Award – $4,000 — March 2026
Gresham Smith Innovation Hackathon

Graduate Fellowship — 2024 - 2029
Carnegie Mellon University College of Engineering

Research Grants

ACCESS – EXPLORE – National Science Foundation — September 2024 – Present

Publications

(Under Review) K. Piper and S. N. Roberts. Beyond Porosity and Permeability: An Alternative Design Space for Passive Capillary Evaporators. September 2026. Preprint available at https://doi.org/10.1184/R1/34814499

(In Preparation) K. Piper, L. T. Biegler, N. Trask, and K. Dayal. Direct Nonlinear Optimization Enables Reduced Order Models for Contact.

K. Piper and V. Agrawal. Ductile fracture in functionally graded materials: Insight into crack behavior within the gradient interface. arXiv preprint arXiv:2411.18642 (2024)

Patents and Technology Development

(Provisional Patent) K. Piper, S. N. Roberts, and T. Daimaru. Novel Geometric Considerations for Solid and Porous Design in Loop Heat Pipe Evaporators. September 2025. *Provisional Patent assigned to California Institute of Technology Sept 2025. Updated August 2026.*

Katherine Piper INV 2026-0097: Design and Manufacturing Processes for High-Volume Production of Two-Phase Loops. *Carnegie Mellon University Invention Discolsure* April 2026.

Katherine Piper, Scott N Roberts. NPO 56342: Novel Geometric Considerations for Solid and Porous Design in Loop Heat Pipe Evaporators. *NASA New Technology Report*, September 2025.

Katherine Piper, Scott N Roberts, Takuro Daimaru. NPO 56343: Hybrid Loop Heat Pipe Architectures. *NASA New Technology Report*, September 2025.

Selected Presentations

(upcoming) Nonlinear Optimization to Exactly Enforce Constraints in Surrogate Modeling. Katherine Piper, *SIAM Conference on Mathematical Aspects of Materials Science*, Invited Presentation, May 2027.

(upcoming) Advancing Diffusion Maps Surrogates with Constrained Optimization. Katherine Piper, *SIAM Conference on Computational Science and Engineering*, Contributed Presentation, February 2027.

Constrained Optimization Methods in Design for Additive Manufacturing. Katherine Piper, *Materials Science and Technology, 2026*, Contributed Presentation, October 2026.

Multiscale Architectured Wick Geometries in Additively Manufactured Loop Heat Pipes. Katherine Piper, *Materials Science and Technology, 2026*, Contributed Presentation, October 2026.

Multiscale LoF Porous Additive Manufacturing for Maximizing Evaporation Rates in Loop Heat Pipes. Katherine Piper, *NextManufacturing Summer Members Meeting, 2026*, Oral Presentation, July 2026.

Fracture in Ductile Functionally Graded Materials. Katherine Piper, *Materials Science and Technology, 2024*, Contributed Presentation, October 2024.

(upcoming) Constrained Optimization of Differential Forms for Learning Multiscale Physics. Katherine Piper, *Frontiers of Mathematical and Computational Aspects of Materials Science*, Poster Presentation, October 2026.

Advancing Diffusion Maps Surrogates with Constrained Optimization. Katherine Piper, *DEVCOM ARL Summer Student Experience Poster Session, 2026*, Poster Presentation, August 2026.

Space Technology, on Earth: A Study of Two-Phase Cooling in Datacenters. Claire Bielski, Katherine Piper, et al., *Carnegie Mellon University Energy Week*, Poster Presentation, March 2026.

Teaching

Carnegie Mellon University | Pittsburgh, Pennsylvania
- Teaching Assistant, Numerical Methods in Engineering — Fall 2024, Fall 2025, Fall 2026
- Eberly Center Future Faculty Program — Spring 2026-Present

California Institute of Technology | Pasadena, California
- Teaching Assistant, Introduction to Mechanics of Materials — Fall 2022, Winter 2023
- Teaching Assistant, Design and Fabrication — Spring 2023, Spring 2024
- Teaching Assistant, Mechanical Prototyping — Summer 2023

Media Coverage and Public Engagement

Students pitch sustainability innovations at inaugural Hackathon — March 4, 2026

Carnegie Mellon University Press Release - Project Featured as First Place Team

Loop Heat Pipes for Smallsat Swarms (LHPss) — October 2024

Scott Roberts: NASA JPL FY24 Progress Report - Named Key Contributor

Other Professional Experience

Special Aerospace Services (Now Aurex Defense) | Arvada, Colorado
- Engineer In Training — August 2020 – August 2021
- Engineering Intern — Summer 2019, Summer 2020

Colorado Space Business Roundtable | Denver, Colorado
- Summer Fellowship Program — Summer 2018